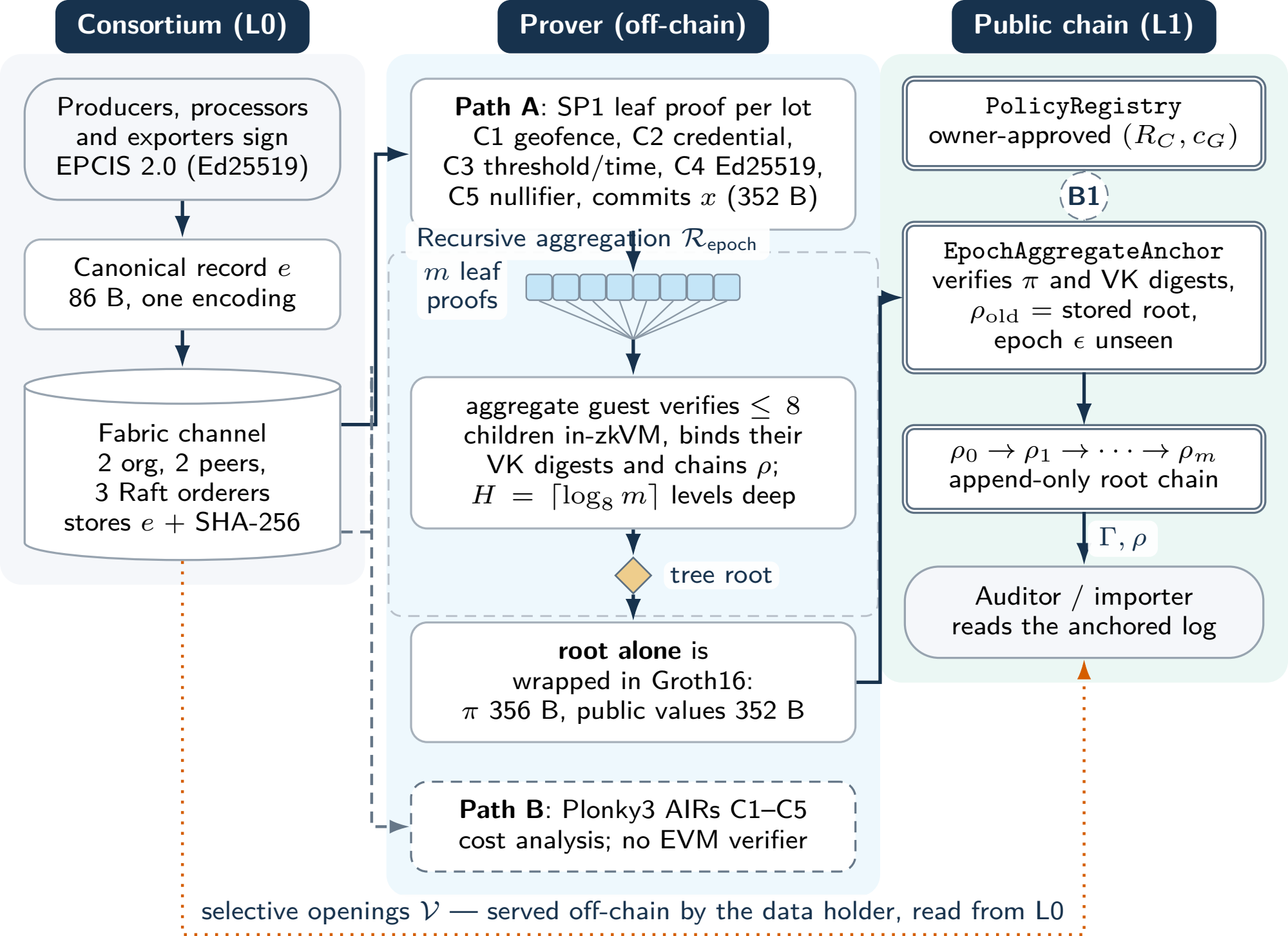

Consortium (L0)
Prover (off-chain)
Public chain (L1)
Producers, processors and exporters sign EPCIS 2.0 (Ed25519)
Canonical record e 86 B, one encoding
Fabric channel 2 org, 2 peers, 3 Raft orderers stores e + SHA-256
Path A: SP1 leaf proof per lot C1 geofence, C2 credential, C3 threshold/time, C4 Ed25519, C5 nullifier, commits x (352 B)
Recursive aggregation $\mathcal{R}_{\text{epoch}}$
m leaf proofs
aggregate guest verifies ≤ 8 children in-zkVM, binds their VK digests and chains ρ; $H = \lceil \log_8 m \rceil$ levels deep
tree root
root alone is wrapped in Groth16: π 356 B, public values 352 B
Path B: Plonky3 AIRs C1–C5 cost analysis; no EVM verifier
PolicyRegistry owner-approved (R_C, c_G)
B1
EpochAggregateAnchor verifies π and VK digests, ρ_{old} = stored root, epoch ϵ unseen
$\rho_0 \to \rho_1 \to \cdots \to \rho_m$ append-only root chain
Γ, ρ
Auditor / importer reads the anchored log
selective openings $\mathcal{V}$ — served off-chain by the data holder, read from L0